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Winding number and homotopy for quaternionic curves

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Abstract

Following a recent approach to quaternionic curves, we defined the quaternionic polar angle that enabled us to define global properties of quaternionic curves, namely the winding number and the homotopy concept. The results admit various applications, including further analogies to plane curves, and physical applications.

keywords: curves in Euclidian space; Hyper-Kähler and quaternionic Kähler geometry; Quaternion and other division algebras.

MSC class. codes: 53A04; 53C26; 11R52.

Contents

1 INTRODUCTION

The study of quaternionic curves is a relatively novel area of research, beginning in [1, 2] and taking impulse in recent times with several interesting results, including theoretical developments [3], applications of various sorts of quaternionic curves [4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14], and the study of evolvents and evolutes [15, 16, 17, 18, 19]. More recently, a novel approach [20] took benefit from the algebraic properties of quaternions to obtain the Frenet-Serret equations, and to determine evolutes and evolvents of quaternionic curves. In this article, the formalism introduced

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in [20] is applied to define the winding numbers of quaternionic curves and to determine several properties, and thus establishing an analogy to the well known subject of curves in $\mathbb{R}^2$ [21], which deserves a section in current textbooks on differential geometry [22, 23, 24].

More specifically, this article reaches several targets. The first one is to generalize the winding number of plane curves in $\mathbb{R}^2$ or $\mathbb{C}$ to quaternionic curves, and to apply it to obtain a quaternionic version for the fundamental theorem of algebra. Despite the previous quaternionic versions of the fundamental theorem of algebra [25, 26], to the best of our knowledge it was not interpreted in terms of winding numbers, and this novel approach is given here. However, the deepest insight that emerges is a method to generalize the complex curves through the transformation of the complex imaginary unit i into the pure imaginary quaternionic function $\omega(t)$, that is a quaternionic curve parametrized by the real variable t. This simple, although not obvious idea, is what proportionate the novel view of quaternionic curves presented in this article.

The notational conventions and the elementary facts about quaternioninc hyper-complexes that will be useful in this article are available in the Section 2, while the main subject of the article, and the novel results are comprised within Sections 3 and 4.

2 QUATERNIONIC BACKGROUND

There are many sources that introduce and describe the quaternions [27, 28, 29, 30], and we only describe the notation for representing quaternions that will be useful in this article, and of course nothing in this section is new. Thus, we initially state that quaternionic numbers ($\mathbb{H}$) are hyper-complexes that, whenever $q\in\mathbb{H}$, then

$$q = x_0 + x_1 i + x_2 j + x_3 k, \qquad \text{where} \qquad x_\mu\in\mathbb{R}, \qquad \text{and} \qquad \mu\in\{0,\,1,\,2,\,3\}. \tag{1}$$

The anti-commuting imaginary units i, j and k comply with

$$i^2 = j^2 = k^2 = -1, \qquad \text{and} \qquad ijk = -1. \tag{2}$$

Likewise the complex numbers, the quaternionic conjugate and the quaternionic norm read

$$\overline{q} = x_0 - x_1 i - x_2 j - x_3 k, \qquad\qquad |q| = \sqrt{q\overline{q}} = \sqrt{x_0^2 + x_1^2 + x_2^2 + x_3^2}\,. \tag{3}$$

Adolf Hurwitz proved that quaternions comprise one of the four division algebras [31], and the other ones are the reals ($\mathbb{R}$), the complexes ($\mathbb{C}$) and the octonions ($\mathbb{O}$). The extended or cartesian notation for quaternions (1) may also be switched to

$$q = x_0 + \omega|\boldsymbol{x}| \qquad \text{where} \qquad \omega = \frac{\boldsymbol{x}}{|\boldsymbol{x}|} \qquad \text{and} \qquad \boldsymbol{x} = x_1 i + x_2 j + x_3 k. \tag{4}$$

This representation encompasses the composed imaginary unit ω, so that $\omega^2=-1$. Moreover, x_0 and $\boldsymbol{x}$ are respectively nominated the scalar (or temporal) and the vector (or spatial) parts of the quaternion. This notation has the opportune feature that the scalar and the vector components are commutative, and the inconvenience that every quaternionic number has its proper imaginary unit ω that neither commutes nor anti-commutes with the imaginary unit of other quaternionic number. The polar notation of the cartesian quaternion is such that

$$q = |q|\big(\cos\theta + \sin\theta\,\omega\big), \qquad \text{where} \qquad \theta\in[0,\,\pi], \tag{5}$$

and the imaginary component of (4) is always positive, and hence the range of the polar angle θ is narrow when compared to the complex numbers. Ultimately, the symplectic notation for quaternions asserts that

$$q = z_0 + z_1 j, \qquad \text{where} \qquad z_0 = x_0 + x_1 i \qquad \text{and} \qquad z_1 = x_2 + x_3 i. \tag{6}$$

The symplectic notation is besides not unique, and (6) is exchangeable with

$$q = z_0 + \overline{\zeta}\,k \qquad \text{where} \qquad \zeta = x_3 + x_2 i, \tag{7}$$

and two supplementary possibilities replacing i with j and k in (6). The polar form displayed in symplectic notation renders

$$q \;=\; |q|\Big(\cos\vartheta e^{i\phi} + \sin\vartheta e^{i\psi} j\Big) \qquad \text{where} \qquad \vartheta \in \Big[0,\,\frac{\pi}{2}\Big] \qquad \text{and} \qquad \phi,\,\psi \in [0,\,2\pi]. \tag{8}$$

The range of the symplectic polar angle ϑ is even more restrictive than (5) thanks to the positive definite character of their trigonometric functions, that are defined in terms of the moduli of the complex components. From reference [32], we adopt the scalar product for quaternions

$$\big\langle p,\,q\big\rangle \;=\; \mathfrak{Re}\big[\,p\overline{q}\,\big], \tag{9}$$

which authorzes us to state that if p and q are orthogonal, then

$$\big\langle p,\,q\big\rangle \;=\; 0, \tag{10}$$

and also that p and q are parallel whenever

$$\big\langle p,\,q\big\rangle \;=\; p\overline{q}. \tag{11}$$

Perceivable, properties (9-11) are valid also for complex numbers, and can be understood as generalizations. The inner product (9) enables us to gain the orthogonality relations, which in cartesian notation are

$$\big\langle q,\,e_\ell q\big\rangle = 0, \qquad \text{where} \qquad e_\ell = \big\{i,\,j,\,k\big\}. \tag{12}$$

Quaternions undoubtedly comprise a four dimensional real vector space, but the polar notations (5) and (6) subsequently give

$$\langle q,\,\omega q\rangle = 0, \qquad \text{and} \qquad \langle q,\,qj\rangle = 0, \tag{13}$$

where the quaternions comprise two dimensional real vector spaces in terms of these notations. An unexpected property, considering that the space has actually four real dimensions from (12). To illuminate this point, let us remember that the restriction of the range of θ in (5) and ϑ in (8) indicates that negative signals ought to be assimilated in the complex structure in order to keep the polar angles within the correct range. In the simplest case, the negative of a cartesian polar quaternion q, we have

$$-\,q = (-1)(\cos\theta + \omega\sin\theta). \tag{14}$$

If $-q$ were complex, the product would signify a rotation of the polar angle, so that $\theta \to \theta+\pi$, something not allowed in the quaternionic case. Hence, the negative signal have to be accommodated within the imaginary unit ω, and the precise rotation of the angle will be $\theta \to \pi-\theta$. Furthermore, this multiplication reverses the orientation of the polar angle, and hence the quaternionic polar angle is not orientable accoding to this operation. A desiring outcome in dimensions higher than two, where the polar angle is supposed to be non-orientable. Using the above precept, we further have

$$\left.\begin{aligned} q \;&=\; \cos\theta + \omega\sin\theta \\ \omega q \;&=\; \cos\left(\theta + \tfrac{\pi}{2}\right) + \omega\sin\left(\theta + \tfrac{\pi}{2}\right) \\ \omega^2 q \;&=\; \cos\left(\pi - \theta\right) + \widetilde{\omega}\sin\left(\pi - \theta\right) \\ \omega^3 q \;&=\; \cos\left(\tfrac{\pi}{2} - \theta\right) + \widetilde{\omega}\sin\left(\tfrac{\pi}{2} - \theta\right) \end{aligned}\right\} \qquad \theta \in \Big[0,\,\frac{\pi}{2}\Big] \qquad \text{and} \qquad \widetilde{\omega} = -\omega. \tag{15}$$